C.2. Unveiling the Impact of Label Noise on Model Calibration in Deep Learning

C.2.1. THE AI SCIENTIST-V2 Idea

Idea

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"Name": "label_noise_calibration",
"Title": "Unveiling the Impact of Label Noise on Model Calibration in Deep Learning",
"Short Hypothesis": "Label noise not only degrades model accuracy but also adversely affects
model calibration and uncertainty estimation; by systematically studying this impact, we can
develop methods to improve both accuracy and calibration under label noise.",
"Related Work": "Previous studies have focused on the impact of label noise on model accuracy
and have proposed methods to mitigate this issue, such as robust loss functions and label
correction techniques. However, there is limited research on how label noise affects model
calibration and uncertainty estimation. For instance, works like 'Dynamics-Aware Loss for
Learning with Label Noise' (Li et al., 2023) address robustness to label noise but do not
explore calibration aspects. Our proposal distinguishes itself by systematically investigating
the effect of label noise on model calibration, which is crucial for reliable deployment of
deep learning models in real-world applications.",
"Abstract": "Label noise is a prevalent issue in real-world datasets, where incorrect
annotations can degrade the performance of deep learning models. While the impact of label
noise on model accuracy has been extensively studied, its effect on model calibration and
uncertainty estimation remains underexplored. Model calibration measures how well the predicted
probabilities reflect the true likelihood of outcomes, which is vital for risk-sensitive
applications that rely on uncertainty estimates for decision-making. In this research, we
propose to systematically investigate how different types and levels of label noise affect the
calibration of deep learning models. We hypothesize that label noise leads to overconfident and
miscalibrated predictions, undermining the reliability of uncertainty estimates. Through
controlled experiments on benchmark datasets with synthetic label noise and real-world datasets
with inherent label noise, we will analyze calibration metrics such as Expected Calibration
Error (ECE) and reliability diagrams. Additionally, we will assess the effectiveness of
existing label noise mitigation techniques in improving model calibration. The findings from
this study will provide insights into the relationship between label noise and model
calibration, guiding the development of more robust models that maintain reliable uncertainty
estimates despite noisy labels.",
"Experiments": [
    "Introduce varying levels and types of synthetic label noise (e.g., symmetric and
    asymmetric noise) into benchmark datasets like CIFAR-10 and MNIST.",
    "Train deep learning models (e.g., ResNet, CNNs) on these noisy datasets and evaluate their
    accuracy and calibration using metrics like ECE and reliability diagrams.",
    "Analyze how different label noise levels impact model calibration compared to their effect
    on accuracy.",
    "Apply existing label noise mitigation techniques, such as robust loss functions and label
    correction methods, to assess their effectiveness in improving calibration.",
    "Evaluate models on real-world datasets known to contain label noise (e.g., web-scraped
    datasets) to validate the findings in practical scenarios.",
    "Conduct ablation studies to understand the interplay between label noise, model
    calibration, and uncertainty estimation."
],
"Risk Factors and Limitations": [
    "Results may be specific to the selected models and datasets, potentially limiting
    generalization to other architectures or domains.",
    "Measuring calibration accurately requires sufficient test data; small test sets may lead
    to unreliable calibration metrics.",
    "Existing mitigation techniques may not significantly improve calibration, indicating a
    need for developing new methods.",
    "Synthetic label noise may not capture all aspects of real-world label noise, affecting the
    applicability of the findings."
],
"Code": "from datasets import load_dataset\nfrom huggingface_hub import ..."
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Link to more material: https://github.com/SakanaAI/AI-Scientist-ICLR2025-Workshop-Experiment/tree/master/label-noise.